



Pharmacological safety evaluation of a traditional herbal medicine “Zereshk-e-Saghir” and assessment of its hepatoprotective effects on carbon tetrachloride induced hepatic damage in rats



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ABSTRACT

Ethnopharmacological relevance: “Zereshk-e-Saghir” (ZES), one of the traditional herbal medicines in old manuscripts of Persian hakims, has been used for the treatment of liver disorders. This current study is aimed to evaluate ZES effects on animal model to investigate its safety and hepatoprotective activity.

Materials and methods: ZES was prepared according to a traditional method by blending aqueous extracts of *Berberis vulgaris* L., with fine particles of other plants including *Rosa damascene* Mill, *Cichorium intybus* L., *Cucumis sativus* L., *Portulaca oleracea* L., *Rheum palmatum* L., and *Nardostachys jatamansi* DC.. The lethality of ZES was determined in male NMRI mice. Acute organ toxicity of ZES (750 and 1500 mg/kg for 15 days, orally) was evaluated by measuring the cell blood count, liver marker enzymes, creatinine, antioxidant status and histopathological examinations in rats. CCl₄-induced liver toxicity was used to examine the hepatoprotective effects of the preparation. The rats were pretreated with 250, 500, 750 and 1500 mg/kg ZES by gavage for 15 days. At day 16, the rats were intraperitoneally injected 1 ml/kg CCl₄ in olive oil. Forty-eight hours after CCl₄ injection, the animals were sacrificed and their liver samples and blood were collected for determination of alanine aminotransferase, aspartate aminotransferase and alkaline phosphatase (ALT, AST, and ALP), histopathological examinations and antioxidant status.

Results: Treatment of the mice with a single dose of ZES up to 2 g/kg did not cause mortality. Treatment of the rats with doses of 750 and 1500 mg/kg for 15 days showed no significant hematotoxicity and hepatotoxicity. Treatment of the rats with ZES reduced the increased serum levels of ALT, AST, and ALP induced by CCl₄ at the doses of 250, 500, and 750 mg/kg. This was almost confirmed by histopathological examinations. Pretreatment with ZES also decreased lipid peroxidation and maintained the levels of glutathione and total antioxidant capacity.

Conclusions: The present in vivo study revealed that the long term usage of ZES was safe for organs in laboratory animals. Meanwhile, prescribing the traditionally-recommended dose of ZES can be probably used against the liver injuries induced by xenobiotics. Further studies in other models of liver injuries are recommended for finding the exact hepatoprotective mechanism of ZES.

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1. Introduction

Detoxification is one of the vital operations of the liver in the human body to clear foreign and naturally occurring substances including chemicals and xenobiotics. This process produces reactive intermediates such as reactive oxygen species (ROSs) which induce oxidative stress and liver injuries (Halliwell, 1991; Wiseman and Halliwell, 1996). Oxidative stress is an imbalance

between prooxidant and antioxidant in the body (Rahal et al., 2014).

Because of production of ROSs and induction of oxidative stress, carbon tetrachloride (CCl_4) induced hepatotoxicity has been a widely used animal model for investigation of liver injuries induced by xenobiotics. Metabolism of CCl_4 by phase I cytochrome P450 produces reactive free radicals, decreases hepatic glutathione content and initiates a chain lipid peroxidation in the hepatocyte membrane (Jaeschke et al., 2002; Recknagel, 1967).

Although there are endogenous defensive antioxidant mechanisms to limit the adverse effects of ROSs, the excessive production of ROSs overwhelms these mechanisms and leads to oxidative stress. Here, the considerable importance of free radical scavengers called antioxidants to protect body is highlighted (Devasagayam et al., 2004; Ulicna et al., 2003). Synthetic antioxidants are cost-effective and efficient, but there are concerns about the safety of these products in long term usage (Brewer, 2011; Pokorny, 2007). So, there have been interests in finding natural antioxidants in recent years. Plants are rich sources of antioxidants whose antioxidant activities could be related to the presence of bioactive constituents such as total polyphenols, total flavonoids, and compounds with Fe^{2+} -chelating activity (Barreira et al., 2008).

Herbal preparations have attracted the attention of clinicians, and more than 700 mono- and poly-herbal preparations have been derived from more than 100 plants in different forms such as decoction, tincture, tablets and capsules for clinical use (Ram, 2001). Although botanical drugs have been iteratively recommended in complementary and alternative medicine for treating the liver disorders, there is not enough evidence for their efficacy and safety recommended by World Health Organization (WHO) for traditional herbal medicines (Stickel and Schuppan, 2007; World Health Organization, 2002).

Traditional Iranian Medicine (TIM) or Persian Medicine dates back to over 8000 years BC, and it is a branch of Arabic Unani medicine with a rich and long history of practicing and studying medicine for diagnosis, prevention and treatment of diseases from ancient times to the present (Jorjani, 2011; Rezaeizadeh et al., 2009). Pharmacotherapy has been playing a key role in this therapeutic system. Several available manuscripts from glorious Persian hakims recommended herbal preparations such as hepatotherapeutic agents (Aghili Khorasani Alavi Shirazi, 2011; Jorjani, 2011). One of these herbal medicines is called “Zereshk-e-Saghir” (ZES). This preparation is a combination of micro fine particles of seven herbal plants as follows: *Berberis vulgaris* L. (Berberidaceae), *Rosa damascena* Mill (Rosaceae), *Cichorium intybus* L. (Compositae), *Cucumis sativus* L. (Cucurbitaceae), *Portulaca oleracea* L. (Portulacaceae), *Rheum palmatum* L. (Polygonaceae) and *Nardostachys jatamansi* DC. (Valerianaceae).

Although the herbal preparation ZES has been used to treat the liver disorders in folk remedy, still there has been no experimental study on its hepatoprotective effects and safety. Therefore, the antioxidant and hepatoprotective effects of ZES on CCl_4 -induced acute liver injury model in rats. Meanwhile, acute organ toxicity of the preparation was studied to determine its safety.

2. Methods and materials

2.1. Chemicals

All the chemicals and reagents used in this study were of analytical grade purity. Carbon tetrachloride, ethylenedinitrilotetraacetic acid, ketamin, xylazine, formalin solution (neutral buffered, 10%), trichloroacetic acid, *n*-butanol, acetic acid glacial, tris-buffered saline, sodium acetate trihydrate, hydrogen chloride,

Table 1

Characteristics of ZES preparation.

Plant	Family	Used part	Amount (% w/w)
<i>Berberis vulgaris</i> L.	Berberidaceae	Fruit	39.2
<i>Rosa damascena</i> Mill	Rosaceae	Petal	19.7
<i>Cichorium intybus</i> L.	Compositae	Seed	11.7
<i>Cucumis sativus</i> L.	Cucurbitaceae	Seed	11.7
<i>Portulaca oleracea</i> L.	Portulacaceae	Seed	11.7
<i>Rheum palmatum</i> L.	Polygonaceae	Rhizome	4
<i>Nardostachys jatamansi</i> DC.	Valerianaceae	Rhizome	2

iron (III) chloride hexahydrate, and iron (II) Sulfate heptahydrate were purchased from Merck Company. Malondialdehyde, 5, 5'-dithio bis-2-nitrobenzoic acid and 2, 4, 6- tripyridyl-s-triazine were obtained from Sigma-Aldrich. Gallic acid and emodin were obtained from Carl Roth Company.

2.2. Herbal preparation

2.2.1. Preparation of “Zereshk-e-Saghir” (ZES)

The characteristics of the herbal preparation, ZES, are shown in Table 1.

Local plants were collected from farms in Kerman province (30.3°N, 57.0°E) from April to August 2013. All the used species were authenticated by an herbalist. Voucher specimens for *Berberis vulgaris* L. (KF1629), *Rheum palmatum* L. (KF2003), *Nardostachys jatamansi* DC. (KF2001), *Rosa damascena* Mill (KF1362), *Cichorium intybus* L. (KF1157), *Cucumis sativus* L. (KF1391) and *Portulaca oleracea* L. (KF1512) were deposited in the Herbarium of Pharmacognosy Department, Kerman University of Medical Sciences, Kerman, Iran. Quality control tests were carried out on the plant samples in accordance to British Pharmacopoeia (2013). Phytochemical analysis by using thin-layer chromatography (TLC) indicated presence of flavonoids including quercetin, kaempferol, and rutin, and anthraquinones (emodin) in the plants which are in consistent with previous studies (Bhandari et al., 2007; Bladt and Zgainski, 1984; Street et al., 2013; Zhou et al., 2015).

For preparation of ZES, all the used parts of plants were powdered by ball-mill[®] and sieved to obtain homogenous particles (mesh no. 30). The obtained powder was mixed with aqueous extract of *B.vulgaris* which was prepared by maceration method and dried with freeze dryer. The proportion of each plant is shown in Table 1. This preparation was made in the Faculty of Pharmacy, Kerman University of Medical Sciences, Kerman, Iran. For pharmacologic studies, ZES was given to animals by mixing with oxy-mel syrup as vehicle according to the traditional recipe. OS was prepared by dissolving 50 g sugar in 100 ml distilled water which was added to 50 ml vinegar. The mixture was boiled to the final volume of 100 ml.

Presence of emodin in methanolic extract of *R. palmatum* L. and ZES was confirmed by TLC (Bladt and Zgainski, 1984). ZES was standardized on the basis of emodin by UV-spectrophotometer at 435 nm (Sucheta et al., 2011) which was equivalent to 432.0 ± 1.1 mg emodin/100 g ZES. Also total phenolic content (Altunkaya and Gokmen, 2008) of ZES was determined by Folin-Ciocalteu reagent and spectrophotometric determination (1232.1 ± 1.6 mg eq. G. A/100 g ZES).

2.2.2. Aqueous herbal extract

For measuring the antioxidant effect of the herbal plants by ferric reducing ability of plasma (FRAP) assay, aqueous extracts were prepared using the following method. The powdered part of the plants were macerated in distilled water (1/5, %w/v) for 6 h (h), and this process was repeated 4 times at room temperature for 24 h. The collected extracts were freeze dried, and finally the

extracts were weighed and kept at -20°C to use later.

2.3. Animals

Male NMRI mice (18–28 g) and male Wistar rats (150–210 g) were purchased from Kerman neuroscience research center, Kerman, Iran. Animals were housed in standard conditions of temperature and humidity with a 12 h light-dark cycle and free access to food and water. All the procedures were carried out according to the Institutional Animal Ethics Committee (IAEC), Kerman University of Medical Sciences, Kerman, Iran.

2.4. Experimental design

2.4.1. Acute toxicity (LD_{50}) determination

The mice were divided into 3 groups each including 10 mice. The first group was treated with normal saline (NS 10 ml/kg; negative control); the second group with ZES 1000 mg/kg; and the third group with ZES 2000 mg/kg. All groups received a single dose of the treatment orally. The animals were evaluated daily for mortality, toxic symptoms, weight, fur conditions, and behavior abnormality.

2.4.2. Evaluation of ZES organ toxicity

For evaluation of organ toxicity of ZES, thirty-two male rats were divided into 4 groups of which the 1st two groups were treated orally with NS and OS (10 ml/kg/day: control groups) for 15 days, and the 2nd two groups with ZES (750 and 1500 mg/kg/day) for 15 days. After sacrificing the rats at day 16, blood and liver samples of the rats were collected for blood cell count (red and white blood cells), liver marker enzymes (ALT, AST, and ALP), creatinine, histopathological examinations and antioxidant status.

2.4.3. CCl_4 -induced liver toxicity in rats

Sixty-four male rats were evaluated for hepatoprotective effects of ZES preparation and were divided into 8 groups. Two of the groups were treated with NS (10 ml/kg, as control), OS (10 ml/kg, as vehicle group), and other 4 groups with ZES (250, 500, 750, and 1500 mg/kg). This range was selected according to the recommended therapeutic dose of ZES in TIM (about 7.5 g/d). The treatment for these groups lasted for 15 days. At day 16, the groups received CCl_4 1 ml/kg (1:1 in olive oil) intraperitoneally (i.p.). In addition, the remaining two groups of the animals only received NS and OS at dose of 10 ml/kg for 15 days. Forty-eight hours after CCl_4 injection, all the 8 groups of the animals were anesthetized by Ketamin/Xylazine, and the blood samples were withdrawn from right atrium with heart puncture for determination of alanine aminotransferase (ALT), aspartate aminotransferase (AST) and alkaline phosphatase (ALP). The liver tissue samples were removed from right parts of the liver lobes and segregated into two parts: one part was fixed in 10% formalin for histopathological examinations and the other part for measurement of lipid peroxidation.

2.5. Biochemical analysis

The EDTA-anticoagulated whole blood of the animals was used for determination of blood cell tests. Meanwhile, the sera of coagulated blood samples were separated at 4000 rpm/10 min/room temperature and used for determination of ALT, AST, ALP and creatinine (Cr) by auto analyzer Hitachi model 902 (Roch, Japan).

2.6. Histopathological examinations

The fixed tissues in 10% buffered phosphate formalin solution were processed by the routine histology procedures, cut into $4\ \mu\text{m}$

thick slices which were stained with hematoxylin and eosin (H&E) (Fischer et al., 2008). Histopathological changes including inflammation, degenerative changes, necrosis and apoptosis (Achliya et al., 2004) were blindly evaluated by a pathologist.

2.7. Measurement of lipid peroxidation

The basis for determination of lipid peroxidation (LPO) is the colorimetric reaction of LPO products such as aldehydes with thiobarbituric acid (TBA) (Ohkawa et al., 1979), but it was used with few modifications in the current study (Mandegary et al., 2012). Briefly, 100 μl of the homogenized liver tissue was added to tubes containing 1 ml of trichloroacetic acid, and after 10 min the tubes were centrifuged for 13 min at 4100 rpm. Later, 1 ml H_2SO_4 0.05% and 800 μl TBA were added to the pellets, then the tubes were stirred and heated in boiling water for 30 min. After cooling, extraction was done by adding 800 μl *n*-butanol and the optical density of the TBA reacted substances (TBARSs) in butanol phase was measured at 532 nm (Mihara and Uchiyama, 1978). The calibration curve of malondialdehyde (MDA) was used to quantify the TBARSs. The data were expressed mM/mg protein.

2.8. Measurement of glutathione level

Assessment of glutathione (GSH) level was done in serum using Elleman reagent. Formations of yellow complexes after oxidation of GSH by 5, 5'-dithio bis-2-nitrobenzoic acid (DTNB) were measured in 412 nm with spectrophotometer (Kebieche et al., 2009; Sedlak and Lindsay, 1968). Briefly, 1 ml of tris-buffered EDTA (TBE) (0.25 M, pH=8.2) was added to 50 μl serum, and the absorbance (A_1) was measured against a blank. After 15 min, 20 μl of DTNB 10 mM was added to the mixture, and the absorbance of the tubes was re-measured (A_2). Negative control tubes containing TBE and DTNB under the same conditions were prepared, and the absorbance was measured (B). Finally, the following equation was used for determination of GSH level. The data were expressed mM.

$$\text{GSH (mM)} = (A_2 - (A_1 - B)) \times 1.57$$

2.9. Determination of antioxidant activity of aqueous extracts and sera by ferric reducing ability of plasma assay

The measurement of antioxidant power of the samples (aqueous extracts and rat sera) was determined by ferric reducing ability of plasma (FRAP) assay with slight modifications (Benzie and Strain, 1996). The basis of this method is ferric to ferrous ion reduction at low pH and formation of a colored ferrous-triaryldiazine complex. The stock solutions contained 300 mM acetate buffer, pH 3.6, 10 mM 2,4,6-tripyridyl-s-triazine (TPTZ) solution in 40 mM HCl, and 20 mM $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ solution. The FRAP reagent was prepared freshly by adding 25 ml acetate buffer, 2.5 ml TPTZ solution and 2.5 ml $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$ solution. The prepared mixture was warmed at 37°C for 5 min before using. Five μl of the tested samples were added to 295 μl of FRAP reagent and reading the absorbance of the blue-colored complexes was taken at 593 nm. The standard curve was linear between 125 and 500 mM $\text{FeSO}_4 \cdot 7\text{H}_2\text{O}$. If the FRAP value measure was over the linear range of the standard curve, additional dilution would have been needed. Results are expressed in mM of sera and per mg extracts.

2.10. Statistical analysis

The statistical analyses were carried out by Microsoft office excel 2007 and SPSS software (version 18.0), and the outcomes

Table 2

The biochemistry, liver function tests and blood parameters of rats treated with ZES for 15 days.

Treatment (dose)	AST (IU/L)	ALT (IU/L)	ALP (IU/L)	Cr (IU/L)	RBC (× 10 ⁶ /L)	WBC (× 10 ³ /L)
NS (10 ml/kg)	201.4 ± 27.9	170.7 ± 33.8	983.6 ± 96.9	0.57 ± 0.04	7.8 ± 0.4	12.1 ± 1.2
OS (10 ml/kg)	175.0 ± 24.1	144.5 ± 34.0	732.0 ± 142.4	0.52 ± 0.08	7.6 ± 1.0	11.9 ± 0.4
ZES (750 mg/kg)	157.5 ± 16.6	107.5 ± 29.6	543.7 ± 153.6	0.51 ± 0.02	6.7 ± 1.3	13.5 ± 3.8
ZES (1500 mg/kg)	168.7 ± 22.2	117.5 ± 23.3	595.0 ± 86.9	0.55 ± 0.04	7.5 ± 1.2	10.7 ± 0.8

Male rats treated with doses of ZES (750, and 1500 mg/kg), OS (Oxymel syrup, 10 ml/kg) and NS (normal saline, 10 ml/kg) for 15 days. The data are presented as Mean ± SD (n=8). ZES: Zereshk-e-Saghir; AST: Aspartate aminotransferase; ALT: Alanine aminotransferase; ALP: Alkaline phosphatase; Cr: Creatinine; RBC: Red blood cell; WBC: White blood cell.

were presented as mean ± SD. One way analysis of variance (ANOVA) followed by post-hoc Tukey tests was performed to analyze the data, *p*-Values < 0.05 were considered as significant.

3. Results

3.1. Evaluation of long term toxicity of ZES

3.1.1. Acute toxicity of ZES

No mortality or morbidity was seen after oral administration of single doses of 1000 and 2000 mg/kg of ZES in mice. Administration of higher doses was not feasible.

3.1.2. Organ toxicity of ZES

The biochemistry, liver function tests and blood parameters of the rats treated with ZES for 15 days are shown in Table 2. The results show no notable changes in the liver enzymes, Cr, RBC and WBC of ZES groups in comparison to the control groups.

The histopathological examinations showed that repeated (15 days) administration of different doses (750, 1500 mg/kg) of ZES did not cause any histological abnormalities in the hepatic cells, central vein, and portal triad in comparison to the control groups receiving NS and OS (Fig. 1 A–D).

3.1.3. Effects of ZES on the oxidative stress parameters

Some of the parameters of oxidative stress including level of

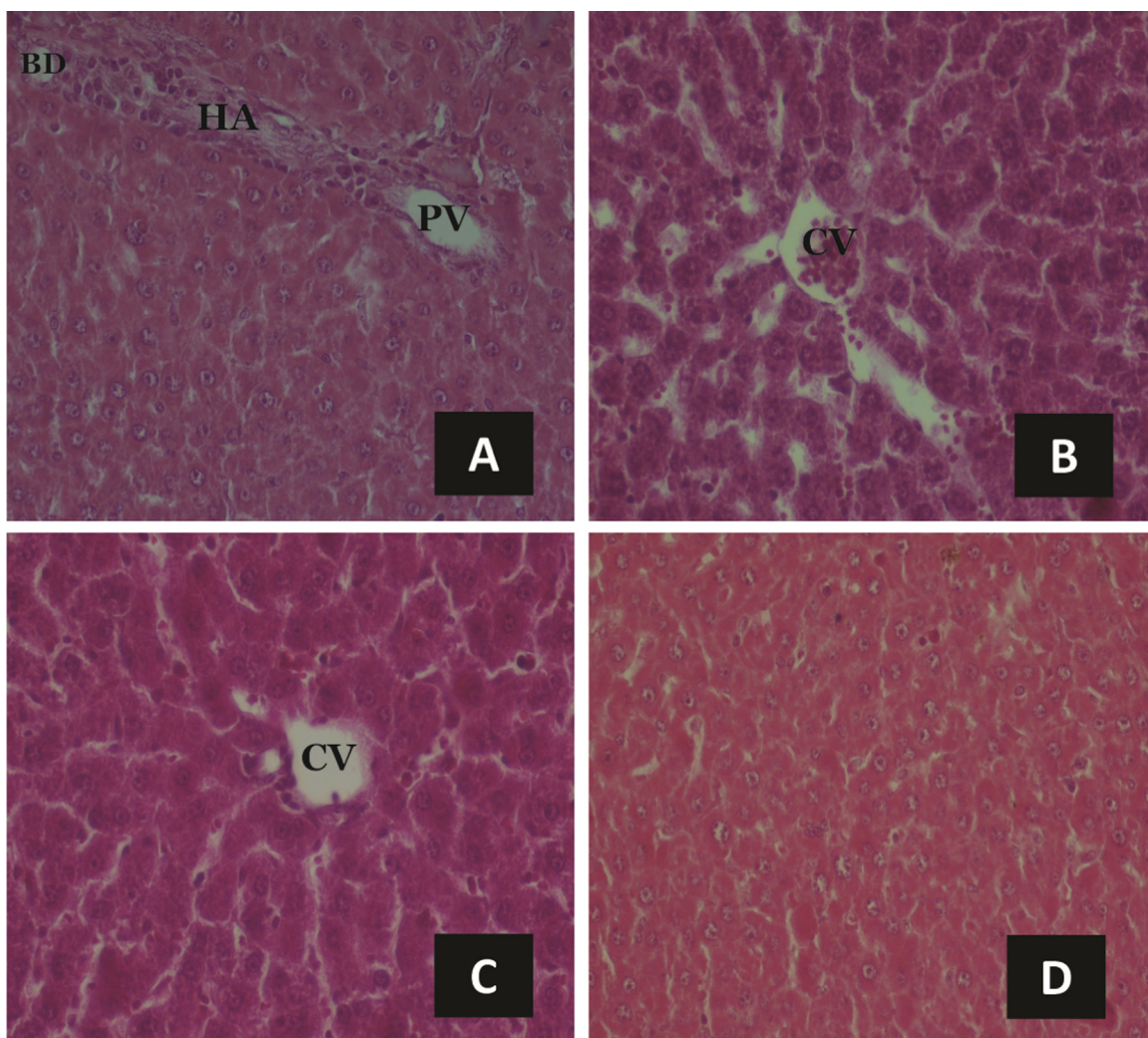


Fig. 1. Effects of ZES on the liver in rats. Rats treated orally with NS 10 ml/kg (A); OS 10 ml/kg (B); ZES 750 mg/kg (C); and ZES 1500 mg/kg (D) for 15 days. The typicality of hepatic cells, central vein (CV), and portal triad (Portal vein (PV)+ Bile duct (BD)+ Hepatic artery (HA)) are observed. ZES: Zereshk-e-Saghir.

Table 3

Effects of the repeated dose of ZES on the oxidative stress parameters (MDA, GSH levels and TAC activities) in rats.

Treatment (dose)	LPO (mM/ mg Protein)	GSH (mM)	TAC (mM)
NS (10 ml/kg)	10.0 ± 1.4	0.2 ± 0.05	0.7 ± 0.08
OS (10 ml/kg)	7.8 ± 1.7	0.2 ± 0.03	0.7 ± 0.2
ZES (750 mg/kg)	6.3 ± 1.3***	0.2 ± .03	1.1 ± 0.2**
ZES (1500 mg/kg)	12.3 ± 2.9	0.1 ± .02	0.9 ± 0.1

Male rats treated with doses of ZES (750, and 1500 mg/kg), OS (Oxymel syrup, 10 ml/kg) and NS (normal saline, 10 ml/kg) for 15 days. The data are presented as Mean ± SD (n=8).

ZES: Zereshk-e-Saghir; LPO: Lipid peroxidation; GSH: Glutathione; TAC: Total antioxidant capacity.

** p < 0.001,

*** p < 0.0001 compared to NS group.

MDA formation as the indicator of LPO, GSH level and total antioxidant capacity (TAC) were measured in the groups that received ZES 750 and 1500 mg/kg for 15 days. According to Table 3, LPO was significantly lower in the group receiving 750 mg/kg of ZES. There were no significant changes in the levels of GSH in different groups. Also ZES at the dose of 750 mg/kg, not 1500 mg/kg, significantly increased the TAC of the serum in rats.

3.2. Evaluation of hepato-protective effects of ZES on CCl₄-induced hepatotoxicity in rats

3.2.1. Establishment of CCl₄-induced liver toxicity in rats

The administration of CCl₄ 1 ml/kg significantly increased the liver enzymes ALT, AST, and ALP in the animals (Fig. 2).

3.2.2. Effects of ZES on the liver enzymes on the CCl₄-induced hepatotoxicity in rats

As it is shown in Table 4, the oral treatment of rats with ZES for 15 days could significantly decrease the levels of liver enzymes in CCl₄-induced hepatotoxicity in the rats.

3.2.3. Effects of ZES on the LPO, GSH and TAC levels in CCl₄-induced

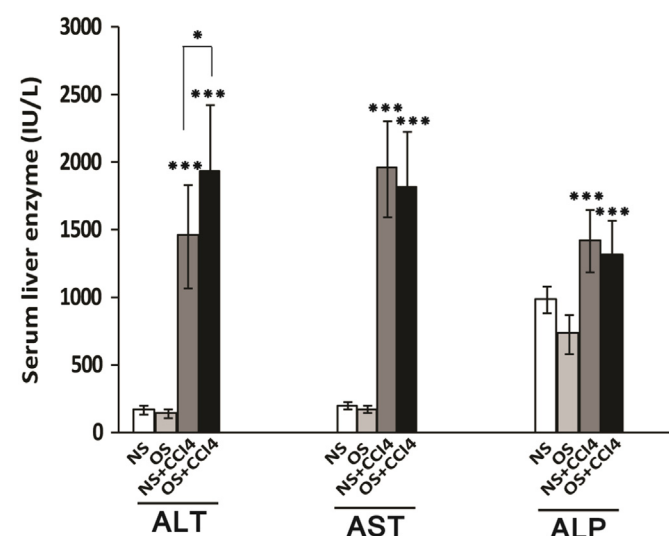


Fig. 2. Levels of liver biomarker enzymes (ALT, AST and ALP) in CCl₄-induced hepatic damage in rats. Male rats received OS (Oxymel syrup, 10 ml/kg) and NS (normal saline, 10 ml/kg) for 15 days before induction of the liver toxicity by single i.p. dose of CCl₄ (1 ml/kg, diluted 1:1 in olive oil). The data are presented as Mean ± SD (n=8). ALT: Alanine aminotransferase; AST: Aspartate aminotransferase; ALP: Alkaline phosphatase. * p < 0.05, ** p < 0.001, and *** p < 0.0001 compared to NS group.

Table 4Effects of ZES on the serum ALT, AST, and ALP levels in CCl₄-induced hepatotoxicity in rats.

Treatment	ALT (IU/L)	AST (IU/L)	ALP (IU/L)
CCl ₄ + NS (1 ml/kg)	1456.4 ± 378.5	1959.9 ± 354.5	1417.1 ± 230.0
OS (10 ml/kg)	144.5 ± 34.0***	175.0 ± 24.1***	732.0 ± 142.4***
CCl ₄ + OS (1 ml/kg)	1938.3 ± 496.7*	1812.2 ± 416.5	1322.2 ± 247.3
CCl ₄ + ZES (250 mg/kg)	1811.1 ± 389.2	1271.1 ± 272.6***	1153.3 ± 91.6
CCl ₄ + ZES (500 mg/kg)	807.5 ± 145.4**	758.2 ± 161.6***	843.7 ± 149.1***
CCl ₄ + ZES (750 mg/kg)	485.6 ± 125.7***	480.0 ± 223.1***	916.2 ± 50.0***
CCl ₄ + ZES (1500 mg/kg)	1348.7 ± 71.9	1308.7 ± 234.5**	1538.7 ± 99.8

Male rats treated with different concentrations of ZES (250, 500, 750, and 1500 mg/kg) and OS (Oxymel syrup, 10 ml/kg) for 15 days before induction of the liver toxicity by a single i.p. dose of CCl₄ (1 ml/kg, diluted 1:1 in olive oil). The data are presented as Mean ± SD (n=8).

ZES: Zereshk-e-Saghir; ALT: Alanine aminotransferase; AST: Aspartate aminotransferase; ALP: Alkaline phosphatase.

* p < 0.05.

** p < 0.001.

*** p < 0.0001 compared to CCl₄ + NS group.

hepatotoxicity in rats

LPO was significantly increased in the liver of the rats treated with CCl₄ (15.3 ± 2.1 vs. 10.0 ± 1.4 in NS group, Fig. 3A). The treatment of the animals by ZES at the doses of 250, 500, 750 mg/kg of ZES could significantly decrease LPO-induced by CCl₄ in the liver of the rats (Fig. 3A).

The treatment with CCl₄ significantly decreased the GSH levels in the serum of the rats (0.09 ± 0.009 mM vs. 0.19 ± 0.05 mM in NS group). The treatment with ZES could significantly increase the GSH levels in the CCl₄-treated rats (Fig. 3B).

The serum TAC of the rats was significantly decreased by giving CCl₄. The treatment with ZES increased TAC at the doses of 250–750 mg/kg in the rats treated with CCl₄ (Fig. 3C).

3.2.4. Effects of ZES on the histopathology of the liver

The effects of ZES on the histopathology of the liver in the CCl₄-induced acute liver toxicity in rats are shown in Fig. 4 and Table 5. The histopathological observations show that CCl₄ induced notable inflammation, degenerative changes, necrosis and apoptosis (Fig. 4. A and B). The treatment of the rats with ZES could decrease the severity of the liver injuries at the doses of 250–750 mg/kg.

3.3. Evaluation of total antioxidant capacity of aqueous extracts of the herbal plants of ZES by ferric reducing ability of plasma (FRAP) assay

Fig. 5 shows the total antioxidant capacity of the ingredients of ZES at concentration of 1 mg/ml. The absorbance of the FRAP solution in the presence of the standard solution of FeSO₄·7H₂O using different concentrations (125–500 mM) was linear with the equation of $y = 0.38x + 0.13$ and $r^2 = 0.998$. The results were adapted for dilution and expressed in mM. The highest antioxidant activities belonged to *Rosa damascena* and *Berberis vulgaris* (the two main ingredients of ZES (59%)).

4. Discussion

Because of the widespread usage of traditional and complementary medicine in some countries, the evaluation of efficacy and safety of these drugs and products is necessary (World Health Organization, 2013). The traditional herbal medicine “Zereshk-e-Saghir” (ZES) is one of the existing prescriptions used frequently by Persian Hakims and Ayurvedic practitioners for treatment of some liver disorders. This preparation has been reported in many old

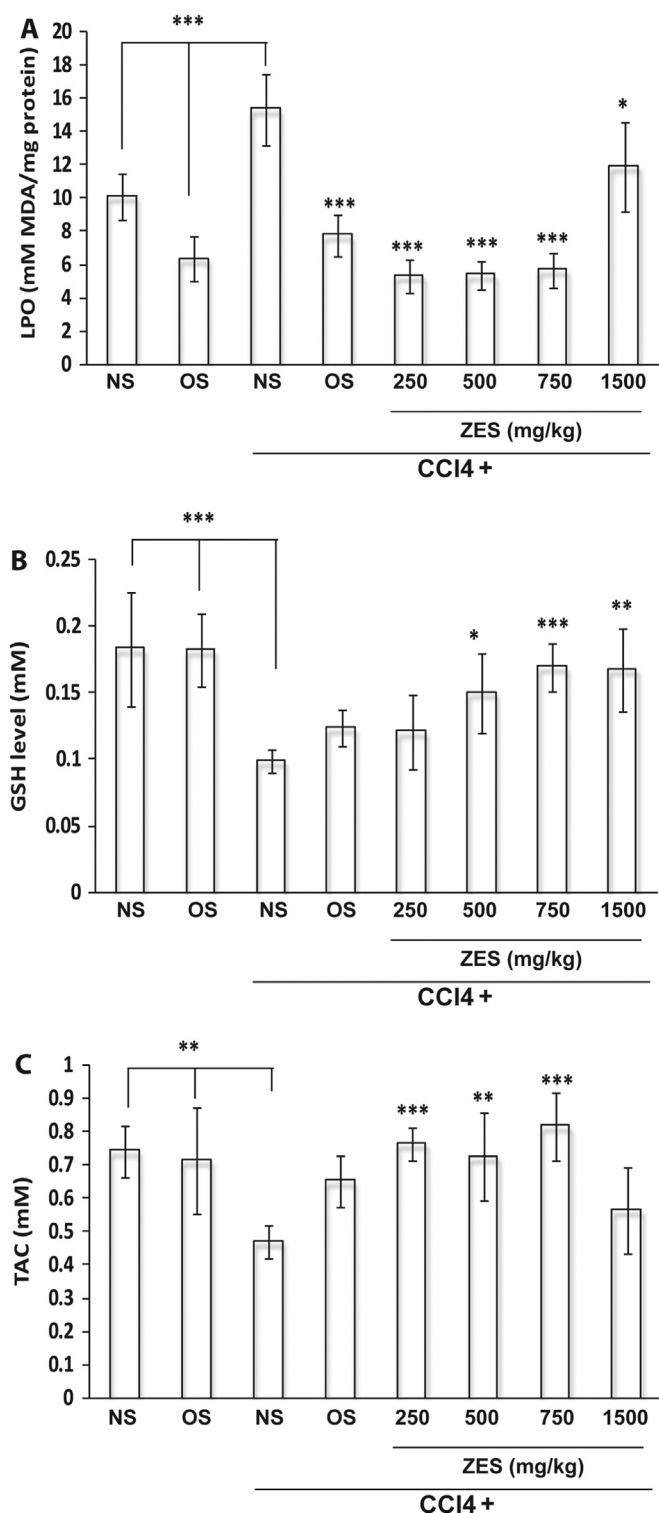


Fig. 3. Effects of different doses of ZES on the levels of LPO (A), GSH (B), and TAC (C) in rats treated with CCl₄. Male rats treated with different concentrations of ZES (250, 500, 750, and 1500 mg/kg), OS (Oxymel syrup, 10 ml/kg) and NS (normal saline, 10 ml/kg) for 15 days before induction of the liver toxicity by single i.p. dose of CCl₄ (1 ml/kg, diluted 1:1 in olive oil). Two control groups of rats treated with NS and OS for 15 days without receiving CCl₄. The data are presented as Mean $\pm$ SD (n=8). ZES: Zereshk-e-Saghir; LPO: Lipid peroxidation; MDA: Malondialdehyde; GSH: Glutathione; TAC: Total antioxidant capacity. * $p < 0.05$, ** $p < 0.001$, and *** $p < 0.0001$ compared to CCl₄+NS group.

pharmacopeias during 1042–1793 CE by glorious Persian Hakims (Aghili Khorasani Alavi Shirazi, 2011; saleh-e ghaeni Heravi, 2012). So, in this study, the safety of ZES and its effects on CCl₄-induced

hepatotoxicity were evaluated in mice and rats.

Oral administration of ZES did not kill any mice at the doses of 1000 and 2000 mg/kg, so the determination of LD₅₀ was not possible. According to these data, doses of 750 and 1500 mg/kg of ZES were administered to the rats for 15 days to evaluate the organ toxicity in a long term usage of ZES. The histopathology of the liver tissues showed no abnormality in the tissue of liver. The evaluation of ZES safety showed that there were no significant changes in the body weight, the level of serum liver enzymes (ALT, AST and ALP) and creatinine. The examination of the blood parameters showed no alterations in the number of RBC and WBC. The determination of the oxidant/antioxidant parameters showed that the treatment with ZES improved TAC and decreased LPO in the liver of the rats at the dose of 750 mg/kg. Interestingly, higher dose of ZES (1500 mg/kg) did not improve the antioxidants capacity of the rats. This effect might be attributed to the hormetic entity of the ZES antioxidants. It is believed that antioxidants such as herbal polyphenols are hormetic; i.e. they have biphasic dose-response relationship, display low-dose beneficial effects and a high-dose toxic effect through generation of reactive and harmful intermediates (Calabrese et al., 2010). This idea is supported by many clinical trials and metabolic studies which have shown no benefit, or even harm, from using antioxidant supplements (Ristow, 2014; Vrolijk et al., 2015).

After the safety evaluation of ZES, the hepatoprotective effects of this preparation were evaluated in CCl₄-induced hepatotoxicity in the rats. CCl₄-induced hepatotoxicity has been widely used to evaluate the hepatoprotective effects of different synthetic and natural compounds. The established mechanism for hepatotoxicity of CCl₄ is microsomal metabolism of CCl₄ which generates of trichloromethyl free radicals ($\cdot\text{CCl}_3$) whose reaction with oxygen forms trichloromethyl peroxy radicals ($\cdot\text{CCl}_3\text{O}_2$) which can lead to LPO and finally cell death (Recknagel, 1967; Recknagel and Ghoshal, 1966). The manifestations of CCl₄-induced hepatotoxicity are fatty liver, cirrhosis and necrosis. Our study showed that the treatment with ZES could significantly decrease the CCl₄-induced liver enzymes markers elevation (ALT, AST and ALP) in the rats. In addition, administration of this herbal preparation caused decline in LPO and replenishment of GSH and TAC. The histopathological observations emphasized these findings. The aqueous extract of *B. vulgaris* was the main part of the preparation (39.2%). Several studies have attributed the antioxidant activities of this fruit to phenolic compounds, vitamin C, malic acid, and tannins (Hanachi and Golkho, 2009; Kiselova et al., 2006; Tomosaka et al., 2008). Also, 20% of ZES was *R. damascena* petals. Some studies have confirmed that *R. damascena* is a source of phenolic compounds with strong in vitro free radical scavenging activity and hepatoprotective effects in vivo (Achuthan et al., 2003; Schiber et al., 2005). Consistent with these evidence, these two aqueous extracts showed the highest antioxidant activity in this study (Fig. 5). The other components of ZES were the powdered seeds of *C. sativus*, *P. oleraceae* and *C. intybus* having an equal ratio (11.7%). A major antioxidant constituent of cucumber, lactic acid, efficiently has scavenged free radicals such as superoxide anion and hydroxyl radicals (Gill et al., 2009; Sotiroudis et al., 2010). Meanwhile there are reports about the anti-diabetic and antioxidant effects of *P. oleraceae* seeds (Sharma et al., 2012; Simopoulos et al., 1992; Yang et al., 2009). The antioxidant activity and hepatoprotective effects of *C. intybus* have been reported by some authors (Ahmed et al., 2003; Heibatollah et al., 2008).

Small amount of rhubarb (*R. palmatum*) rhizome (3.9%) is one of the ingredients of ZES. It has been documented that rhubarb has both therapeutic and toxic effects on the liver. Even though there are reports about the antioxidant and hepatoprotective effects of this plant in low doses (Arosio et al., 2000; Wang et al., 2011), the high doses of rhubarb caused hepatotoxicity (Wang et al., 2011).

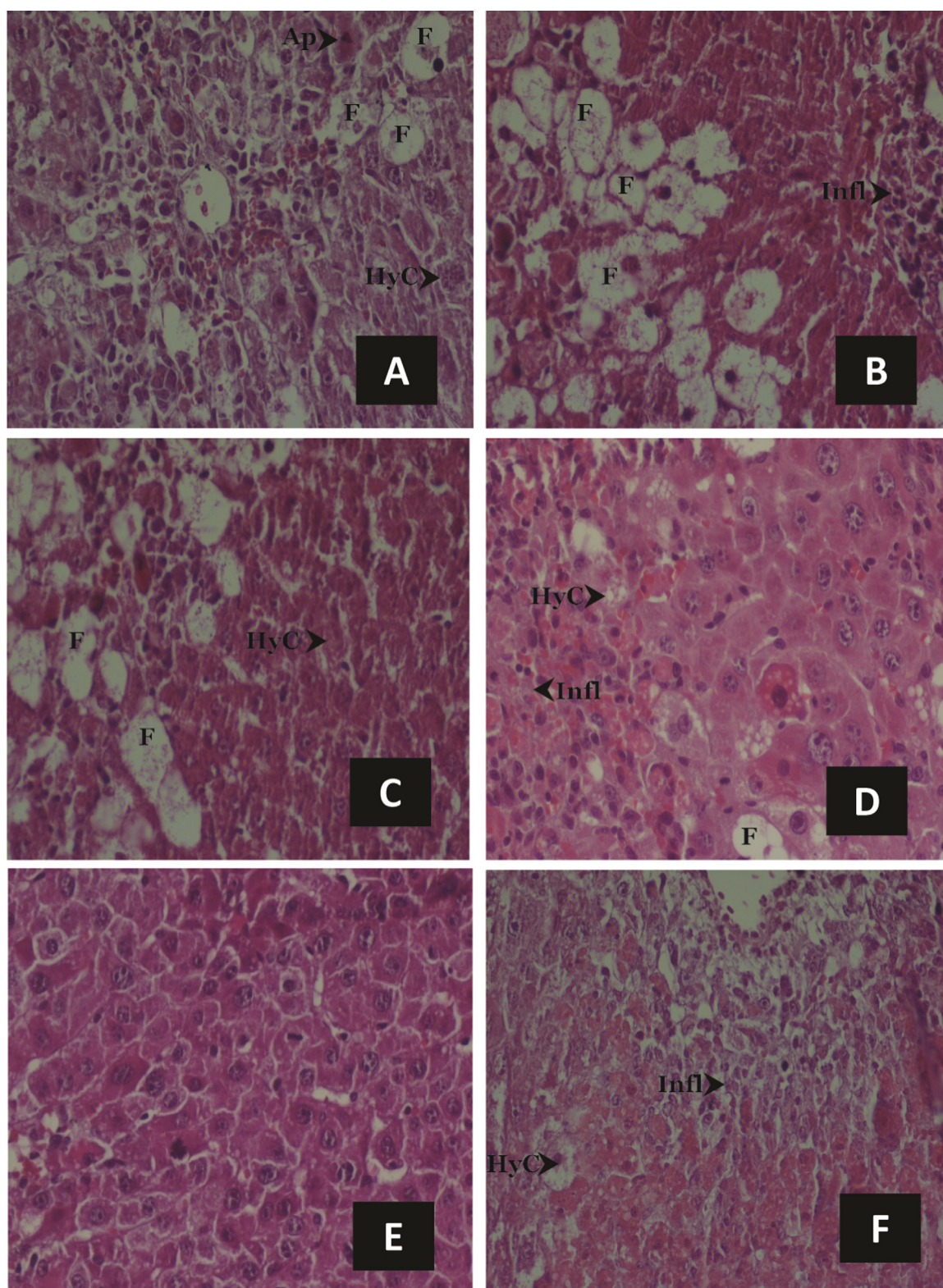


Fig. 4. Effects of ZES on the CCl_4 -induced liver injury in rats. Rats pre-treated orally with NS 10 ml/kg (A); OS 10 ml/kg (B); ZES 250 mg/kg (C); ZES 500 mg/kg (D); ZES 750 mg/kg (E); and ZES 1500 mg/kg (F) for 15 days. The liver injury was induced in rats by i.p. injection of 1 ml/kg CCl_4 in olive oil (1:1 ratio). Hydropic changes (Hyc), fatty accumulation (F), apoptosis (Ap) and inflammatory cells (Infl) are seen in the CCl_4 -induced liver injury. The severities of changes were reduced by ZES pre-treatment. All the histological examinations were performed under a light microscope ($\times 400$ magnifications).

This toxicity might be justified by the hormetic entity of the plant antioxidants including emodin. Emodin is an anthraquinone with both prooxidant and antioxidant properties (Tian and Hua, 2005), and some organ toxicity including hepatitis and nephritis has been

caused by that (Yan et al., 2006). Keeping this in mind, the mild toxicity and lower hepatoprotective effects of higher doses of ZES (> 750 mg/kg) might be assigned to the presence of hormetic antioxidants in the ZES. The lowest component of ZES preparation

Table 5
Effects of ZES on CCl₄-induced liver injury in rats.

Treatment (dose)	Inflammation/ Site of inflammation	Degenerative changes		Necrosis	Apoptosis
		Hydropic changes	Fatty changes		
CCl ₄ + NS (1 ml/kg)	+++ / zone 1 and 2	++	+++	++	++
CCl ₄ + OS (1 ml/kg)	+++ / zone 1 and 2	++	+++	++	++
CCl ₄ + ZES (250 mg/kg)	+ / zone 1	+	+	++	+
CCl ₄ + ZES (500 mg/kg)	+ / zone 1 and 2	+	++	–	+
CCl ₄ + ZES (750 mg/kg)	–	–	–	–	+
CCl ₄ + ZES (1500 mg/kg)	++ / zone 1 and 2	+	++	+	+

NS; Normal saline; OS; Oxymel syrup; ZES; Zereshk-e-Saghir
–: negative; +: mild; ++: moderate; +++: severe

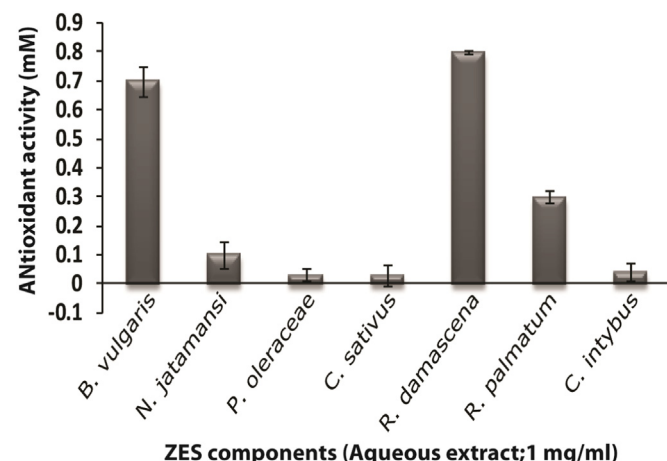


Fig. 5. Antioxidant activities of 1 mg/ml aqueous extracts of ZES components. The antioxidant activities of the extracts were determined using FRAP assay. The data are presented as Mean $\pm$ SD (n=3). ZES: Zereshk-e-Saghir.

was fine particles of rhizomes of *N. jatamansi* (2%). The hepatoprotective activities of *N. jatamansi*, which have been reported in some studies, have been assigned to the antioxidant effects of this plant (Fischer et al., 2008; Subashini et al., 2006).

5. Conclusions

In conclusion, the results of this study showed that “Zereshk-e-Saghir”, a traditional herbal medicine, would be safe for clinical application, and it would be effective for the prevention of xenobiotics induced hepatic injuries. The hepatoprotective effects of ZES might be in part mediated through the antioxidant activities of ZES ingredients. Adherence to the traditionally-recommended ZES regimen is highly recommended.

Authors' contributions

ZS carried out all the experiments, analyzed data and was involved in drafting the manuscript. FSh conceived the study, participated in its design and contributed to writing the manuscript draft. AP also conceived the study and supervised the formulation preparation. SSM and MHN were involved in the animal studies. AM conceived and designed the study, was involved in the data analysis and interpretation, and wrote the manuscript.

Conflict of interests

Authors declare no conflict of interests.

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